

Task 1: Formulating Your World as an MDP

Recall-E Assignments — Week 1

Scenario: Student Evening Decision MDP

We model a student's evening routine as a Markov Decision Process. Each evening the student must decide how to spend their time. Their decision affects their academic performance and well-being over time. The goal is to find a policy that maximises long-term cumulative reward.

1. Formal MDP Definition

1.1 State Space S

The state captures the student's current academic and energy level:

State	Label	Description
s0	Behind	Low academic progress, low energy
s1	On Track	Moderate academic progress, moderate energy
s2	Ahead	Good academic progress, high energy

1.2 Action Space A

Three actions are available from every state:

Action	Label	Description
a0	Study	Spend evening studying / doing assignments
a1	Sleep	Rest early to recover energy
a2	Hang Out	Socialise with friends, leisure activities

1.3 Transition Probabilities $P(s' | s, a)$

Transitions are stochastic — e.g., studying hard might still leave you 'On Track' rather than 'Ahead' if the material was difficult. Key transitions (non-zero entries only):

From s	Action a	To s'	$P(s' s,a)$
Behind (s0)	Study (a0)	Behind (s0)	0.2
Behind (s0)	Study (a0)	On Track (s1)	0.7
Behind (s0)	Study (a0)	Ahead (s2)	0.1

On Track (s1)	Study (a0)	Behind (s0)	0.0
On Track (s1)	Study (a0)	On Track (s1)	0.3
On Track (s1)	Study (a0)	Ahead (s2)	0.7
Ahead (s2)	Study (a0)	On Track (s1)	0.1
Ahead (s2)	Study (a0)	Ahead (s2)	0.9
Behind (s0)	Sleep (a1)	Behind (s0)	0.6
Behind (s0)	Sleep (a1)	On Track (s1)	0.4
On Track (s1)	Sleep (a1)	Behind (s0)	0.1
On Track (s1)	Sleep (a1)	On Track (s1)	0.6
On Track (s1)	Sleep (a1)	Ahead (s2)	0.3
Ahead (s2)	Sleep (a1)	On Track (s1)	0.2
Ahead (s2)	Sleep (a1)	Ahead (s2)	0.8
Behind (s0)	Hang Out (a2)	Behind (s0)	0.8
Behind (s0)	Hang Out (a2)	On Track (s1)	0.2
On Track (s1)	Hang Out (a2)	Behind (s0)	0.4
On Track (s1)	Hang Out (a2)	On Track (s1)	0.5
On Track (s1)	Hang Out (a2)	Ahead (s2)	0.1
Ahead (s2)	Hang Out (a2)	On Track (s1)	0.5
Ahead (s2)	Hang Out (a2)	Ahead (s2)	0.5

1.4 Reward Function $R(s, a)$

Rewards reflect immediate academic and well-being outcomes. Studying from 'Behind' gives a moderate boost; studying when already 'Ahead' yields a smaller marginal gain. Hanging out provides short-term happiness but incurs opportunity cost.

State \ Action	Study (a0)	Sleep (a1)	Hang Out (a2)
Behind (s0)	+3	+1	-1
	+5	+2	+1
Ahead (s2)	+2	+3	+2

2. Value Iteration by Hand ($\gamma = 0.9$)

We perform exactly two steps of Value Iteration on the 3-state MDP defined above. The update rule is:

$$V_{\{k+1\}}(s) = \max_a [R(s, a) + \gamma * \sum_{s'} P(s'|s, a) * V_k(s')]$$

Discount factor: $\gamma = 0.9$

Initialisation — Step 0

$V_0(s) = 0$ for all s :

$$V_0(s_0) = 0 \quad V_0(s_1) = 0 \quad V_0(s_2) = 0$$

Step 1 — Compute $V_1(s)$

Since $V_0 = 0$ everywhere, the expectation term vanishes and $V_1(s) = \max_a R(s, a)$.

$$V_1(s_0): \max(R(s_0, \text{Study})=3, R(s_0, \text{Sleep})=1, R(s_0, \text{HangOut})=-1) = 3 \quad [\text{Study}]$$

$$V_1(s_1): \max(R(s_1, \text{Study})=5, R(s_1, \text{Sleep})=2, R(s_1, \text{HangOut})=1) = 5 \quad [\text{Study}]$$

$$V_1(s_2): \max(R(s_2, \text{Study})=2, R(s_2, \text{Sleep})=3, R(s_2, \text{HangOut})=2) = 3 \quad [\text{Sleep}]$$

State	Best Action	$V_1(s)$
s_0 — Behind	Study	3
s_1 — On Track	Study	5
s_2 — Ahead	Sleep	3

Step 2 — Compute $V_2(s)$

Now V_1 is non-zero. We evaluate $Q(s,a) = R(s,a) + 0.9 * \text{SUM } P(s'|s,a)*V_1(s')$ for every (state, action) pair and take the max.

— $V_2(s_0)$ —

$$Q(s_0, \text{Study}) = 3 + 0.9 * [0.2*3 + 0.7*5 + 0.1*3] = 3 + 0.9 * [0.6 + 3.5 + 0.3] = 3 + 0.9 * 4.4 = 3 + 3.96 = 6.96$$

$$Q(s_0, \text{Sleep}) = 1 + 0.9 * [0.6*3 + 0.4*5 + 0.0*3] = 1 + 0.9 * [1.8 + 2.0] = 1 + 0.9 * 3.8 = 1 + 3.42 = 4.42$$

$$Q(s_0, \text{HangOut}) = -1 + 0.9 * [0.8*3 + 0.2*5 + 0.0*3] = -1 + 0.9 * [2.4 + 1.0] = -1 + 0.9 * 3.4 = -1 + 3.06 = 2.06$$

$$V_2(s_0) = \max(6.96, 4.42, 2.06) = 6.96 \quad [\text{Study}]$$

— $V_2(s_1)$ —

$$Q(s_1, \text{Study}) = 5 + 0.9 * [0.0*3 + 0.3*5 + 0.7*3] = 5 + 0.9 * [1.5 + 2.1] = 5 + 0.9 * 3.6 = 5 + 3.24 = 8.24$$

$$Q(s_1, \text{Sleep}) = 2 + 0.9 * [0.1*3 + 0.6*5 + 0.3*3] = 2 + 0.9 * [0.3 + 3.0 + 0.9] = 2 + 0.9 * 4.2 = 2 + 3.78 = 5.78$$

$$Q(s_1, \text{HangOut}) = 1 + 0.9 * [0.4*3 + 0.5*5 + 0.1*3] = 1 + 0.9 * [1.2 + 2.5 + 0.3] = 1 + 0.9 * 4.0 = 1 + 3.60 = 4.60$$

$$V_2(s_1) = \max(8.24, 5.78, 4.60) = 8.24 \text{ [Study]}$$

— $V_2(s_2)$ —

$$Q(s_2, \text{Study}) = 2 + 0.9 * [0.0*3 + 0.1*5 + 0.9*3] = 2 + 0.9 * [0.5 + 2.7] = 2 + 0.9 * 3.2 = 2 + 2.88 = 4.88$$

$$Q(s_2, \text{Sleep}) = 3 + 0.9 * [0.0*3 + 0.2*5 + 0.8*3] = 3 + 0.9 * [1.0 + 2.4] = 3 + 0.9 * 3.4 = 3 + 3.06 = 6.06$$

$$Q(s_2, \text{HangOut}) = 2 + 0.9 * [0.0*3 + 0.5*5 + 0.5*3] = 2 + 0.9 * [2.5 + 1.5] = 2 + 0.9 * 4.0 = 2 + 3.60 = 5.60$$

$$V_2(s_2) = \max(4.88, 6.06, 5.60) = 6.06 \text{ [Sleep]}$$

Summary of Value Iteration Results

State	V_0	V_1	V_2	Greedy Policy
s0 — Behind	0	3	6.96	Study
s1 — On Track	0	5	8.24	Study
s2 — Ahead	0	3	6.06	Sleep

3. Interpretation

After two steps of Value Iteration, the greedy policy recommends Studying from states s0 (Behind) and s1 (On Track), and Sleeping from s2 (Ahead). This is intuitive: studying yields the highest immediate reward and moves the agent towards better states when behind; once ahead, rest preserves that advantage. With more iterations and larger gamma, the value estimates will converge to the true optimal value function, and the policy may refine further.